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EEG analysis in Attention-Deficit/Hyperactivity Disorder: a comparative study of two subtypes

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Abstract

This study investigated differences in the EEG between children with Attention-Deficit/Hyperactivity Disorder of the Combined Type, Attention-Deficit/Hyperactivity Disorder of the Predominantly Inattentive Type and control subjects. All subjects were between the ages of 8 and 12 years, and groups were matched on age and gender. The EEG was recorded during an eyes-closed resting condition from 21 monopolar derivations and these were clustered into nine regions prior to analysis. One minute of trace was analysed using Fourier transformation to obtain both absolute and relative power estimates in the delta, theta, alpha and beta frequency bands. The patient groups were found to have greater levels of theta and deficiencies of alpha and beta in comparison to the control group. Children with Attention-Deficit/Hyperactivity Disorder of the Predominantly Inattentive type were found to be significantly different from those of the Combined type in the same measures, appearing to be closer to the normal profiles. The general results support a maturational lag model of the central nervous system in Attention Deficit/Hyperactivity Disorder. The differences between the subtypes suggest a difference in the severity of the disorder rather than a different neurological dysfunction.

Keywords: Attention Deficit/Hyperactivity Disorder subtypes; Children; EEG; Maturation; Diagnosis

1. Introduction

Over the course of this century the disorder that has become known as Attention Deficit/Hyperactivity Disorder (ADHD) has undergone clarification in its aetiology. Initially ADHD was believed to result from brain damage, but this explanation lost favour as children without brain damage were diagnosed with ADHD. Subsequently, researchers in the 1950s and 1960s changed the name of this disorder from ‘minimal brain damage’ to ‘minimal brain dysfunction’

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(Green and Chee, 1994). In 1968 the DSM-II first listed diagnostic criteria for ADHD under the title 'hyperkinetic reaction of childhood', characterised by overactivity, restlessness, distractibility and short attention span (APA, 1968). Much of the literature from the 1960s and 1970s used both 'minimal brain disfunction' (MBD) and 'hyperactive' to describe the same disorder. In DSM-III (APA, 1980), the title was changed to 'Attention Deficit Disorder' and two groups were identified, children with and without hyperactivity. In the DSM-IV (APA, 1994), the diagnostic criteria have changed again, with three main groups being identified: ADHD of the Predominantly Hyperactive-Impulsive Type, ADHD of the Predominantly Inattentive Type (ADHDin) and ADHD of the Combined Type (ADHDcom). This disorder is primarily found in boys (James and Taylor, 1990), with the ratio of boys to girls being approx. 4:1 for all three DSM-IV groups (De Quiros et al., 1994).

With normal maturation, EEG frequencies increase as a function of age, with slow wave activity apparently being replaced by faster waveforms (Matousek and Petersen, 1973; Matthis and Scheffner, 1980). John et al. (1980) developed 32 linear regression equations predicting the frequency composition of the EEG as a function of age. The results indicated that development of the normal EEG was linear in nature. Benninger et al. (1984), in a longitudinal study of 96 boys and girls, found that theta activity decreased as alpha increased and that the speed of change in occipital areas was almost twice that of central areas. Gasser et al. (1988a) found that certain regions of the brain matured before other regions. Absolute power in delta, theta and alpha 1 frequency bands was found to decrease and amplitudes to become similar with age. The decline was found to be greatest in posterior regions. Frontally, delta and theta were found to develop in parallel, whereas theta dominated delta in all other areas. Alpha activity showed a strong posterior increase. At frontal and central regions, the increase started later and remained small. All beta activity showed a decline with age. Except for alpha 2 activity, all frequency bands and total power showed a continuous decrease in power with age. For relative power, a strong complementary replacement of theta by alpha 2 activity was found up to the age of 14. Delta, theta and alpha 1 frequencies decreased with age and higher frequencies increased. All of these studies found a decrease in slow wave activity and an increase in faster frequency bands with age, with this change being linear in nature.

Topographic studies of maturation have found changes to take place from posterior to anterior regions. Gasser et al. (1988b) found that delta, theta and alpha waves were developed earliest occipitally followed by parietal, central and frontal regions. Beta waves developed earliest in central regions followed by parietal, occipital and then frontal regions. In the central area, the midline was found to have a lower frequency power than the two hemispheres, whereas high frequency power was found more evenly distributed between the three regions.

Electrophysiological studies of children with learning and behavioural problems have found that these children have differences in the EEG when compared to normal control subjects (John et al., 1988). Studies of mentally retarded children (Gasser et al., 1983a), learning disabled children (Lubar et al., 1985) and hyperactive children (Capute et al., 1968; Wikler et al., 1970) have found an increase in slow wave activity in the EEG.

Satterfield et al. (1973a), with a group of good responders to stimulant medication, found an increase in slow wave activity and greater power in the lower frequency bands between 0 and 8 Hz, prior to medication being prescribed. Matousek et al. (1984) found the highest correlates of MBD in the relative delta band for parieto–occipital derivations. Mann et al. (1992), in a study of children with ADHD, found an increase in absolute amplitude in the theta band during a resting condition, predominantly in the frontal regions. During cognitive tasks, ADHD children showed a greater increase in theta activity in frontal and central regions, and a decrease in beta activity in posterior and temporal regions, with tasks requiring sustained attention. The ADHD children were found to have EEG frequency distributions that resembled profiles typical of younger children. Mann et al. (1992) concluded that this finding supported the view that ADHD reflects maturational delays in the systems that subserve attention.

The studies outlined above have led researchers to propose two models of ADHD with a neurophysiological basis. The first model proposes that ADHD is the result of a low level of central nervous system (CNS) arousal (Satterfield and Cantwell, 1974). To investigate this, Satterfield and Dawson (1971) conducted a study of skin conductance levels (SCLs). This study found that 50% of the hyperactive children had abnormally low SCLs, supporting a model of low CNS arousal. Hyperactive children have been found to respond well to the use of stimulant medication and are often prescribed stimulants as part of their clinical management (Spencer et al., 1996). Two of the drugs most commonly used in the treatment of ADHD in Australia are dexamphetamine and Ritalin (Serfontein, 1991). Both of these drugs act as CNS stimulants and have the effect of increasing concentration and reducing excessive motor activity. These findings further support a model of low CNS arousal.

A second model of ADHD, although not incompatible with a low CNS arousal model, is the maturational lag model. Studies of the auditory evoked potential have found that hyperactive children have significantly lower amplitudes and longer latencies than age-matched control subjects (Satterfield et al., 1973a,b). These findings are typical of younger children.

Most studies of ADHD have used populations of children exhibiting hyperactive behaviours with few studies investigating subtypes of ADHD. Kuperman et al. (1996), using the DSM-III-R criteria, studied quantitative EEG differences between children with ADHD, Undifferentiated Attention Deficit Disorder (UADD) and normal children. For relative power, main effects of band were found, with the control group having more delta than the UADD subjects and less beta than both groups of children with ADD. Only the UADD group had hemispheric differences, with decreased delta and increased beta in the left hemisphere. This study investigated topographic effects only where a significant main effect was found for the band and this may have resulted in significant regional differences being overlooked. An eyes-open condition was also used, making comparison with most other EEG studies of ADHD difficult.

Chabot and Serfontein (1996) studied EEG differences in children with ADD with or without hyperactivity, using the DSM-III criteria. The children with ADD were found to have an increase in absolute and relative theta, with the greatest increase being found in frontal regions and at the midline. A slight elevation in relative alpha was also noted, and in a small group of subjects (13%), an increase in beta was found. The differences between the two ADD groups mainly involved degree of abnormality, not type. From these results, Chabot and Serfontein (1996) concluded that the EEG patterns represented a deviation from normal development, not a maturational lag.

DSM-IV (APA, 1994) outlined diagnostic criteria for three subgroups of children with ADHD. This study investigated two of these subgroups of children with ADHD: children with ADHDcom and ADHDin, to determine whether the two subgroups differ from normal control subjects in their EEG. A further aim of this study was to investigate EEG differences between the ADHDcom and the ADHDin subgroups to determine if the subgroups were neurologically independent. It is anticipated from the literature cited that differences between the control group and the two ADHD groups will be found in the theta band frontally and the delta and beta bands in posterior regions.

2. Method

2.1. Subjects

Three groups of 20 children, with 16 boys and four girls in each group, participated in this study, with the gender ratio of 4:1 to represent the ratio of boys to girls often found for this disorder. All children were between the ages of 8 and 12 years and right handed and footed. For inclusion, all subjects were assessed using the Wechsler Intelligence Scale for Children (WISC-III) and were required to have a full scale score of 85 or higher.

The groups used were children diagnosed with ADHDcom or ADHDin and a control group. Both groups of children with ADHD were drawn from new patients referred by their family doctor to a Sydney-based paediatric practice for an initial assessment for ADHD. No subjects had a history of medication use for ADHD. The control group came from local schools and community groups.

Inclusion in the ADHD groups was based on a clinical assessment by a paediatrician and a psychologist. Diagnostic criteria used were from the DSM-IV for both ADHDcom and ADHDin. Assessment was based on a clinical history from a parent, school reports for a minimum period of the previous 12 months, reports from any other health professionals and behavioural observations made during the assessment. Inclusion as a control subject was based on a structured interview with a parent. Subjects who scored outside the normal age range on the Neale Analysis of Reading or below the average range on the Wide Range Achievement Test (WRAT) spelling were excluded from the control group. The CPRS-48 Connors rating scale was used, with subjects who scored above a *T*-score of 65 on any of the six measures being excluded.

Inclusion in all groups was also based on the following criteria: an uneventful prenatal, perinatal and neonatal period; no disorders of consciousness, head injury with cerebral symptoms, history of central nervous system diseases, obvious somatic diseases, convulsions, history of convulsive disorders, paroxysmal headache, enuresis or encopresis after the fourth birthday, tics, stuttering, obvious mental diseases; and no deviation with regard to mental and physical development. Any children who showed signs of depression, anxiety, oppositional behaviour or syndromal disorders were excluded from this study.

Children in all groups were excluded if atypical spike wave activity was present in the EEG. Across all groups, subjects were matched in 1-year-age bands and by gender.

2.2. Procedure

All subjects were tested in a single session lasting approx. 2.5 h. Subjects were first assessed by a paediatrician, using a structured interview, when a clinical history and physical examination were undertaken. Subjects then had a psychometric assessment by a psychologist. This involved the administration of a WISC-III, Neale analysis of reading and WRAT spelling. At the conclusion of this assessment, subjects had a neurological assessment consisting of evoked potentials and an EEG. The EEG was recorded using an eyes-closed resting condition, with subjects seated on a reclining chair. Recording of the EEG was stopped for a break if subjects became restless or drowsy. Electrode placement was in accordance with the international 10–20 system, using an electrocap produced by Electrocap International. The activity in 21 derivations was divided into nine regions by averaging in each region. These regions were the left frontal (Fp1, F3, F7), midline frontal (Fpz, Fz), right frontal (Fp2, F4, F8), left central (T3, C3), midline central (Cz), right central (T4, C4), left posterior (T5, P3, O1), midline posterior (Pz, Oz) and right posterior (T6, P4, O2). A single electro-oculogram (EOG) electrode referenced to Fpz was placed beside the right eye and a ground lead was placed on the left cheek. Linked ear references were used with all EEG, and reference and ground leads were 9-mm tin disk electrodes.

The EEG was recorded and Fourier transformed by a Cadwell Spectrum 32, software version 4.22, using test type EEG, montage Q-EEG. The sensitivity was set at 150 μV/cm, low frequency filter 0.53 Hz, high frequency filter 70 Hz and 50 Hz notch filter. Reject level for the EOG was set at 50 μV. The sampling rate of the EEG was 200 Hz and the Fourier transformation used 2.5-s epochs.

Thirty 2.5-s epochs were selected from the trace at random and recorded using both visual and computer rejects. These were further reviewed and cut to 24 epochs for analysis. The EEG was analysed in four frequency bands: δ (0.5–2.5 Hz), θ (2.5–7.5 Hz), α (7.5–13.5 Hz) and β (13.5–20.5 Hz).

2.3. Statistical analysis

Analysis of variance was performed examining the effects of region and group for each band in

Table 1
Mean ages and IQ scores for the ADHDcom, ADHDin and control groups

Mean	Control group	ADHDcom group	ADHDin group
Age (months)	122.1	122.7	123.6
Full scale IQ (WISC-III)	116.5	98.8	100.9

absolute and relative power and for the total power. The effects of region were examined in two orthogonal three-level repeated-measures factors. The first of these was a sagittal factor, within which planned contrasts compared the frontal regions with the posterior regions, and their mean with the central regions. The second factor was laterality, within which planned contrasts compared activity in the left hemisphere with that in the right hemisphere, and their mean with the midline region. These planned contrasts allow optimal clarification of site effects within the regions studied. Within the group factor, planned contrasts compared the patient groups with the control group (to establish ADHD differences from normal subjects) and the ADHDcom group with the ADHDin group. Only between-group effects and interactions are reported here for space reasons. All F-values reported have 1 and 53 $(N-1)$ d.f.

3. Results

No significant age differences were found between groups (see Table 1). The control group had a significantly higher mean IQ than the ADHD groups ($F = 29.68$, $P < 0.001$), which did not differ on IQ.

Four subjects in the ADHDcom group were found to be statistical outliers with beta levels greater than 3 S.D. above the mean. These subjects were excluded from further analysis.

Table 2 provides an overview of significant between-group comparisons. Fig. 1 shows the results for total power. The ADHD groups had a greater difference than the control subjects in their midline vs. left/right activity at the frontal compared with posterior regions ($F = 4.13$, $P < 0.05$), and at the central region compared with the frontal and posterior regions ($F = 5.02$, $P < 0.05$), suggesting a largely frontal midline enhancement in total activity in ADHD.

Table 2
Summary of significant comparisons between groups

Comparison	Absolute power					Relative power			
	Total	δ	θ	α	β	δ	θ	α	β
Control vs. ADHD									
Main effect			*				***	***	***
Main effect × F vs. P				*	*	*		*	***
Main effect × F/P vs. C				*				*	
Main effect × L/R vs. M		*	*					*	*
Main effect × F vs. P ×	*		*						
L/R vs. M	*		**						*
Main effect × F/P vs. C × L/R vs. M									
ADHDcom vs. ADHDin									
Main effect							***	*	**
Main effect × F vs. P						*			

Notes. $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$; vs., versus; F, frontal; P, posterior; C, central; L, left hemisphere, R, right hemisphere; M, midline; F/P, mean of the combined frontal and posterior regions; L/R, mean of the combined left hemisphere and right hemisphere regions.

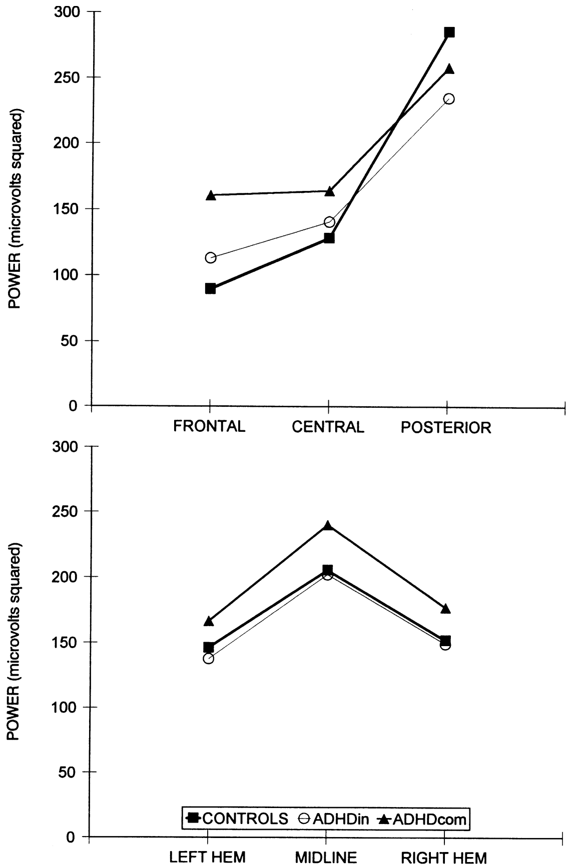

Fig. 1. Total power as a function of scalp distribution: top — sagittal section; bottom — lateral section.

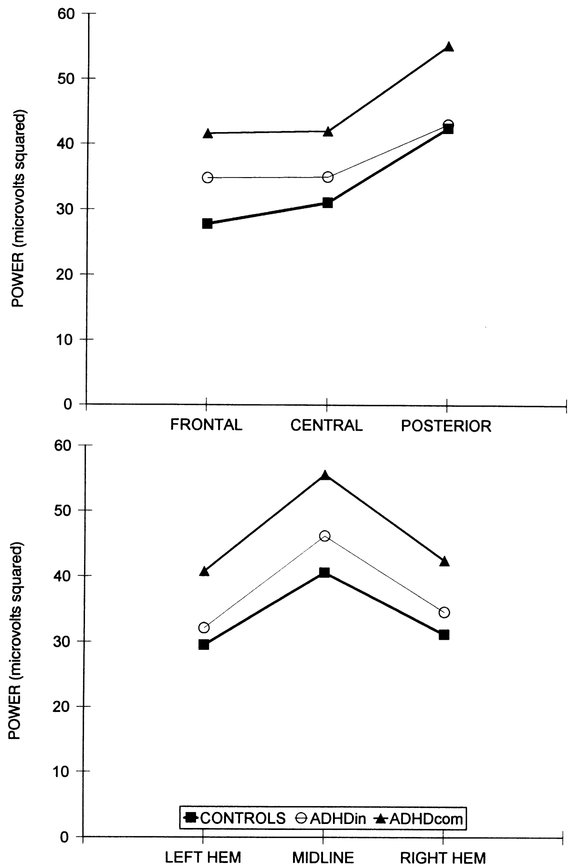

Fig. 2. Absolute delta as a function of scalp distribution, from frontal to posterior (top) and left to right (bottom).

In the absolute delta band (Fig. 2), the difference between the midline and the left and right hemispheres was greater in the ADHD groups than in the control group ($F = 4.44$, $P < 0.05$). For relative delta, the ADHD groups had more posterior (compared with frontal) activity than the control group ($F = 5.90$, $P < 0.05$) and this effect was greater in the ADHDcom group than the ADHDin group ($F = 4.66$, $P < 0.05$).

The distribution of absolute theta is shown in Fig. 3. The ADHD groups had greater theta power than the control subjects ($F = 5.63$, $P < 0.05$). The ADHDcom group had greater theta power than the ADHDin group, a difference approaching significance ($F = 3.85$, $P = 0.055$). The difference between the midline and the two hemispheres was significantly greater in the ADHD groups than the control group ($F = 5.00$, $P < 0.05$). This interaction was greater at the frontal regions than the posterior regions ($F = 6.05$, $P < 0.05$), and greater at central regions than either frontal or posterior regions ($F = 7.71$, $P < 0.01$), indicating a fronto–central maxima for this interaction. The difference between the midline and the two hemispheres was greater at frontal regions than posterior regions in the ADHDcom group compared to the ADHDin group, an interaction approaching significance ($F = 3.87$, $P = 0.054$). Similar results were found in the relative theta band, with significant differences in level between the ADHD groups and the control group ($F = 46.7$, $P < 0.001$), and between the ADHDcom and

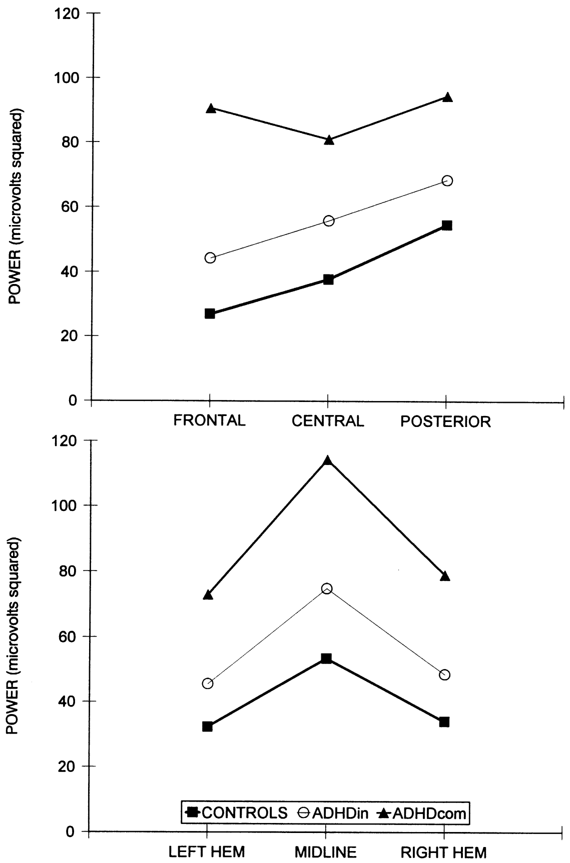

Fig. 3. Absolute theta as a function of scalp distribution, sagittally (top) and laterally (bottom).

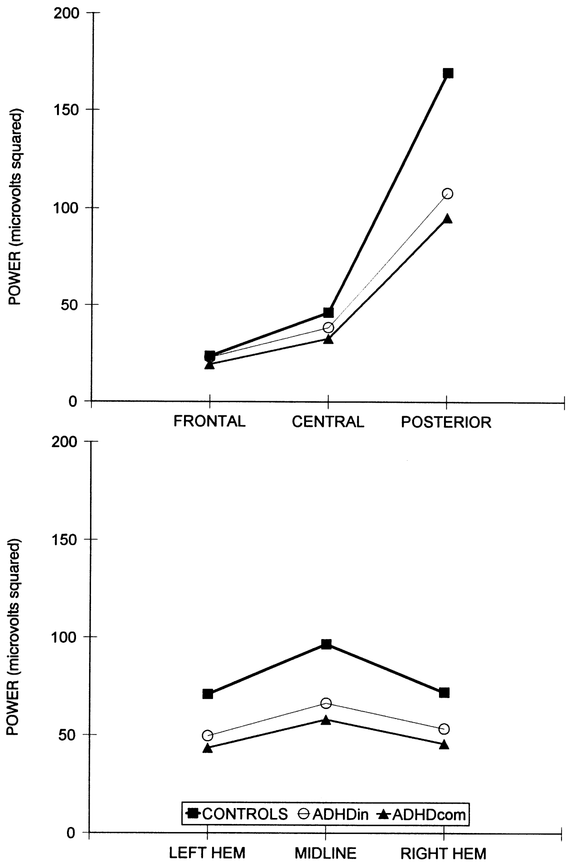

Fig. 4. Absolute alpha as a function of scalp distribution, from anterior to posterior (top) and by hemisphere (bottom).

ADHDin groups ($F = 14.79$, $P < 0.001$). An effect of laterality approached significance, with the ADHDcom group having a greater difference between the midline and the two hemispheres at frontal compared with posterior regions than the ADHDin group ($F = 3.90$, $P = 0.053$).

In the absolute alpha band (Fig. 4), a main effect approaching significance was found, with the ADHD groups having less absolute alpha than the control group ($F = 3.93$, $P = 0.053$). The control group had relatively greater alpha power in posterior than frontal regions compared to the ADHD groups ($F = 5.12$, $P < 0.05$). The difference between the central regions and the mean of the frontal and posterior regions was also greater in control subjects ($F = 6.23$, $P < 0.05$). For relative alpha, the control group had more activity than the ADHD groups ($F = 23.72$, $P < 0.001$), and the ADHDin group had more alpha than the ADHDcom group ($F = 5.49$, $P < 0.05$). The control group had a greater enhancement of relative alpha at posterior regions compared with the ADHD groups ($F = 4.42$, $P < 0.05$), and this difference between groups was least at the central regions compared with frontal/posterior regions ($F = 5.65$, $P < 0.05$). A significant difference occurred between the control group and the ADHD groups in relative alpha at the midline compared with the two hemispheres, with the control group showing the largest effect ($F = 4.82$, $P < 0.05$).

Patterns across regions for absolute beta are presented in Fig. 5. The control group had somewhat more beta than the ADHD groups ($F = 4.00$, $P = 0.051$). This difference was greater in the posterior than the frontal regions ($F = 5.29$, $P < 0.05$). For relative beta, the control group had more power than the ADHD groups ($F = 15.26$, $P < 0.001$), and the ADHDin group had more power than the ADHDcom group ($F = 7.57$, $P < 0.01$). The control group and the ADHD groups differed for the sagittal effect, with the greatest difference between groups occurring at frontal regions ($F = 19.49$, $P < 0.001$). With laterality, the difference between the midline and the two hemispheres was greater in the control subjects than in the ADHD groups ($F = 6.19$, $P < 0.05$). The midline enhancement at the central regions was greater than at the frontal and posterior regions in the control group compared to the ADHD groups ($F = 5.33$, $P < 0.05$).

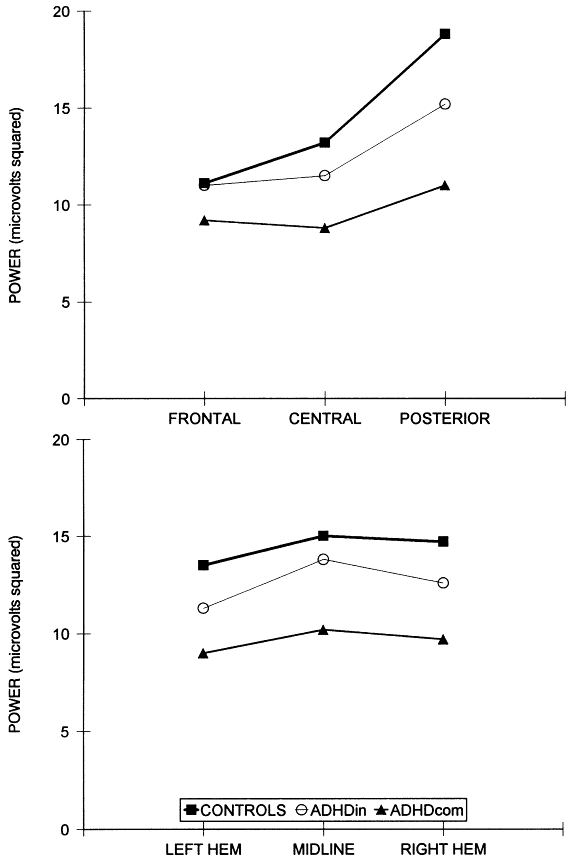

Fig. 5. Absolute beta as a function of scalp distribution, from anterior to posterior (top) and laterally (bottom).

4. Discussion

A significant difference was found between the IQ of the control group and the two ADHD groups. Previous studies of ADHD have reported similar differences, but this has not been considered likely to have affected the EEG results (e.g. Satterfield et al., 1972). Chabot and Serfontein (1996) compared two groups of children with ADD, one group with normal IQs and the other with a low IQ. They concluded that the greatest variance in EEGs was related to the diagnosis and not IQ. The IQ difference between the control group and the clinical groups was possibly exacerbated here by the inclusion criteria for control groups of average performance in reading and spelling. Studies of EEG and IQ have generally been inconclusive (Bosaeus et al., 1977), but Gasser et al. (1983b) found small correlations between increasing IQ scores and measures on the EEG. It is thus possible that some of the between-group differences reflect differences in IQ. The importance of this possibility is reduced by the observation of substantial differences between the two ADHD groups, which did not differ on IQ.

Previous studies of hyperactive children have found an increase in slow wave activity and an increase in amplitude in the 0–8-Hz frequency range in the EEG compared with normal control subjects (Capute et al., 1968; Satterfield et al., 1973, 1974). Similar findings were obtained in this study, with the ADHD subjects having greater levels of both absolute and relative theta over all regions. Most previous studies have found an increase in theta primarily located in the frontal derivations (Mann et al., 1992; Chabot and Serfontein, 1996). However, this study found significantly more frontal midline theta than control subjects. The ADHD groups here had a significant increase in relative delta in the posterior regions, confirming the finding of Matousek et al. (1984) that the greatest correlation of MBD and EEG abnormalities occurred in the relative delta band at posterior sites. Mann et al. (1992) reported that subjects with ADHD had a decrease in absolute amplitude of the beta band in posterior regions. This study confirms the results of Mann et al. (1992) for absolute beta, with the ADHD groups having less posterior beta than the control group. Chabot and Serfontein (1996) found an increase in alpha activity in children with ADD. This was not found in the present study, where the ADHD groups had decreased relative alpha across all sites, with the greatest difference occurring in the posterior regions.

Gasser et al. (1988a) found a systematic decrease in slow wave activity and a complementary increase in faster frequencies with increasing age. This was greatest in the theta and alpha bands, with a high correlation between the decrease in theta and the increase in alpha activity. In the delta band, the strongest decrease with maturation was found in posterior regions. The present findings support a maturational lag model of the CNS in ADHD. The increased posterior delta, increases in theta and a reduction in the alpha bands are all indicative of a maturational lag.

Significant differences across all regions were found between patient groups in the relative theta and alpha bands. ADHDcom subjects had a greater percentage of theta than ADHDin subjects for all regions. In the alpha band, the reverse was found, with the ADHDin subjects having greater alpha than the ADHDcom subjects. A similar finding for relative delta in posterior regions was noted. ADHDcom subjects had a greater percentage of delta than the ADHDin subjects.

This study demonstrated the existence of measurable differences between children with ADHDcom and ADHDin. These differences appear to be in the degree of severity of the differences from the control group, rather than in the nature of the EEG abnormalities, suggesting that these two subgroups of ADHD are not neurologically independent. This is compatible with the results of Chabot and Serfontein (1996).

The principal difference in the diagnosis of the two subtypes of ADHD involves behavioural aspects. Children with ADHDcom are often restless and fidgety, are impulsive, have a short concentration span and are easily distracted. Children with ADHDin have the concentration problems but do not have the gross behavioural aspects of ADHDcom. This study found that ADHDin subjects exhibited EEG abnormalities indicative of a maturational lag, but that the lag was not so great as in those subjects with ADHDcom. These data suggest that the more behavioural problems that are exhibited by a child with ADHD, the greater is the level of the associated EEG abnormalities. This finding is consistent with other studies (e.g. Matousek et al., 1984) that report an increase in EEG abnormalities correlating highly with an increase in the level of abnormality exhibited by the child.

If ADHD were the result of a maturational lag, then a decrease in the percentage of slow wave EEG activity with increasing age would be expected to be found in ADHD. Wikler et al. (1970) noted that the abnormally slow EEG activity of hyperactive children did not increase or decrease with age. Chabot and Serfontein (1996) felt that the EEG abnormalities found were indicative of a deviation from normal maturation, not the result of a maturational lag. Katada et al. (1981), in a study of mentally retarded children, found that the EEG did mature in the same direction as normal children, although the process was delayed. The study of age factors in the development of children with ADHD could provide valuable information relative to the nature of the maturational lag.

The major problem with the maturational lag model is the occurrence of ADHD in adults. The maturational lag model implies that as a child with ADHD gets older, the symptoms of ADHD should lessen. Studies of ADHD have estimated that between 30 and 70% of children with ADHD continue to show symptoms of the disorder as adults (Bellak and Black, 1992). The DSM-IV (APA, 1994), states that some of the behavioural problems, such as excessive gross motor activity, become less in adolescence and adulthood. It is thus possible that maturational lag underlies some components of ADHD but not the entire disorder. Further research is needed to investigate this problem.

An increase in midline theta, compared to the two hemispheres, was found in the frontal regions, with the ADHDcom group having more theta than the ADHDin group. Yamaguchi (1994) found frontal midline theta spread to the frontal half of the scalp and rarely to posterior regions, findings considered to represent an EEG entity associated with concentration. Frontal midline theta was not found in all subjects, although those exhibiting it were found to have a more extroverted personality type. As such an increase in theta was found only in the ADHDcom group, it is possible that it corresponds to a specific behavioural component of ADHD unique to ADHDcom. The frontal and central midline regions were the only regions where the ADHD groups had greater total power than the control group. The overt behavioural problems exhibited by children with ADHDcom may be primarily related to the functioning of the frontal regions of the brain.

Four individuals in the ADHDcom group were excluded from analysis because they had much higher levels of beta than was typical of the entire group. This EEG profile has only been noted once before in children with ADHD (Chabot and Serfontein, 1996), accounting for approx. 13% of their sample. This subset of subjects in two studies, although small in number, suggests the existence of a subtype of ADHDcom, characterised by an increase in beta activity, and this needs further investigation.

This study obtained general results similar to other studies (e.g. Mann et al., 1992; Chabot and Serfontein, 1996), with ADHD subjects having increased levels of theta wave activity, especially in frontal regions, and a decrease in posterior beta. Additionally, a decrease in relative alpha was noted in the present study. Differences in the EEG were also found between the ADHDcom group and the ADHDin group in the theta, alpha and beta bands. This study has demonstrated that EEG can be used to differentiate normal children and children with ADHD, as well as subtypes of the disorder. These data suggest that the subtypes differ in severity rather than in the nature of the underlying neurological impairment. Our findings confirm that EEG may be useful in the clinical assessment of ADHD. Further research is needed in relation to the subtypes of ADHD and aspects of comorbidity within the population, as well as to investigate the reliability of these results for clinical use.

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